

Special Seminar on Multi-Modal AI

Jiayuan Mao

Department of CIS, GRASP

<https://cis7000-fall26.thelabone.org/>

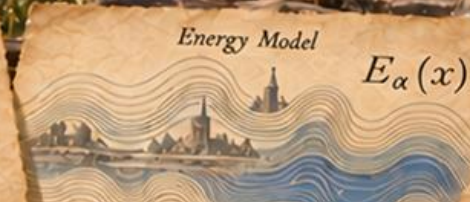
Lecture 6: Text-to-Image Generation

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Initial Proposal

Deadline: This Sunday! Each project has ≤ 4 students

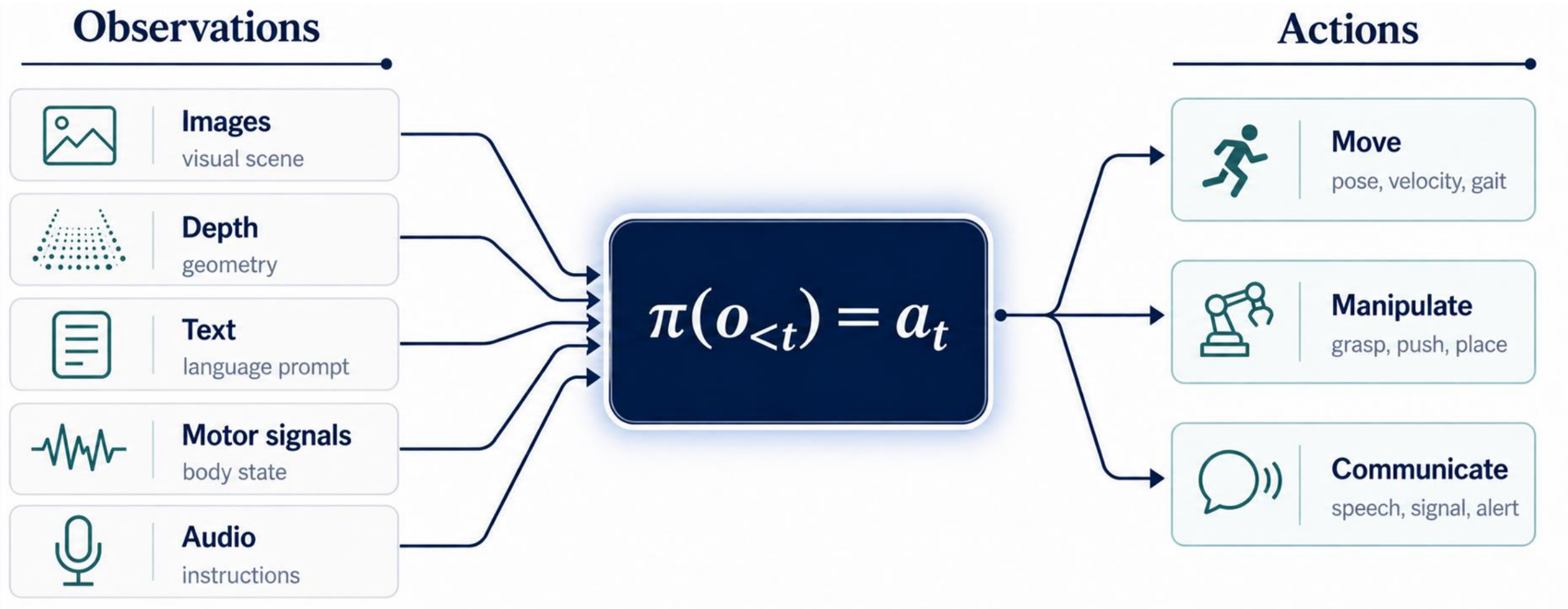
This is initial proposal: I will provide feedbacks

Goal of This Course

Use Machine Learning to Build Intelligent Agents

Abstract View of An Agent: Multimodal AI

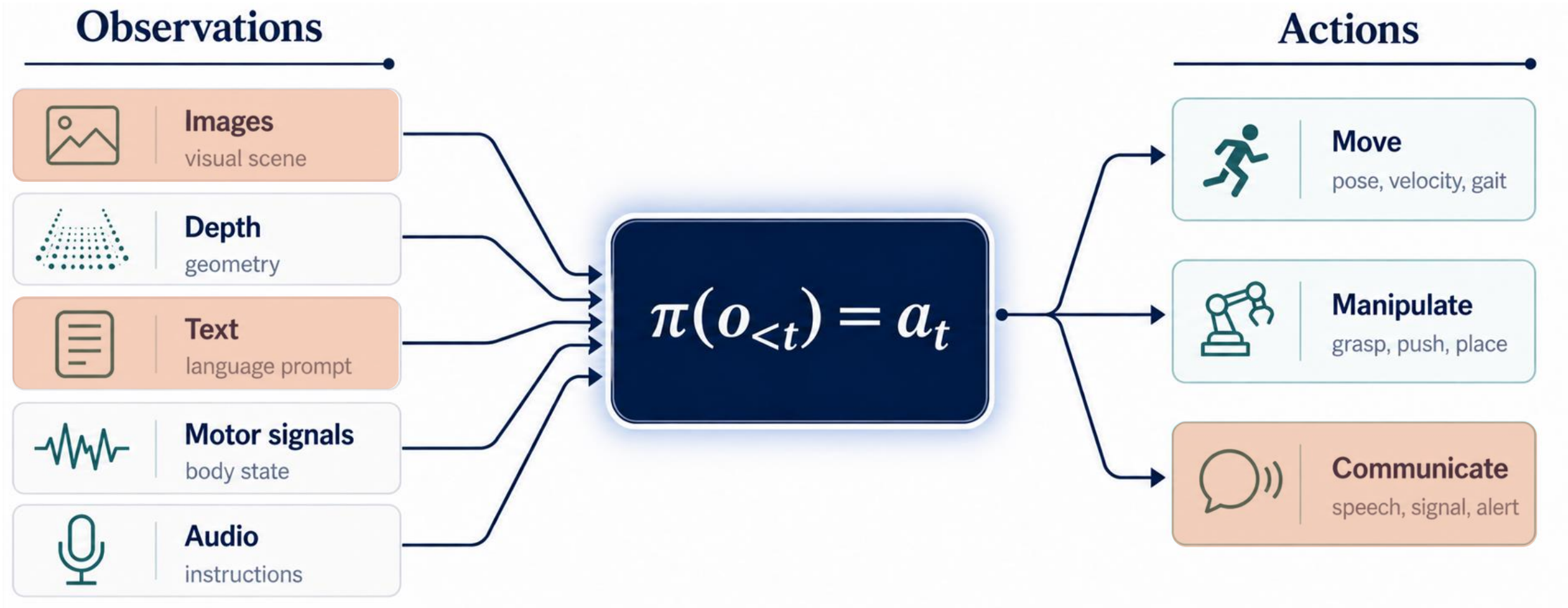
π : observations $\rightarrow$ action



Notation: t – time. $o_{<t}$ - all observations before t . a_t - action at t .

Abstract View of An Agent: Multimodal AI

π : observations $\rightarrow$ action



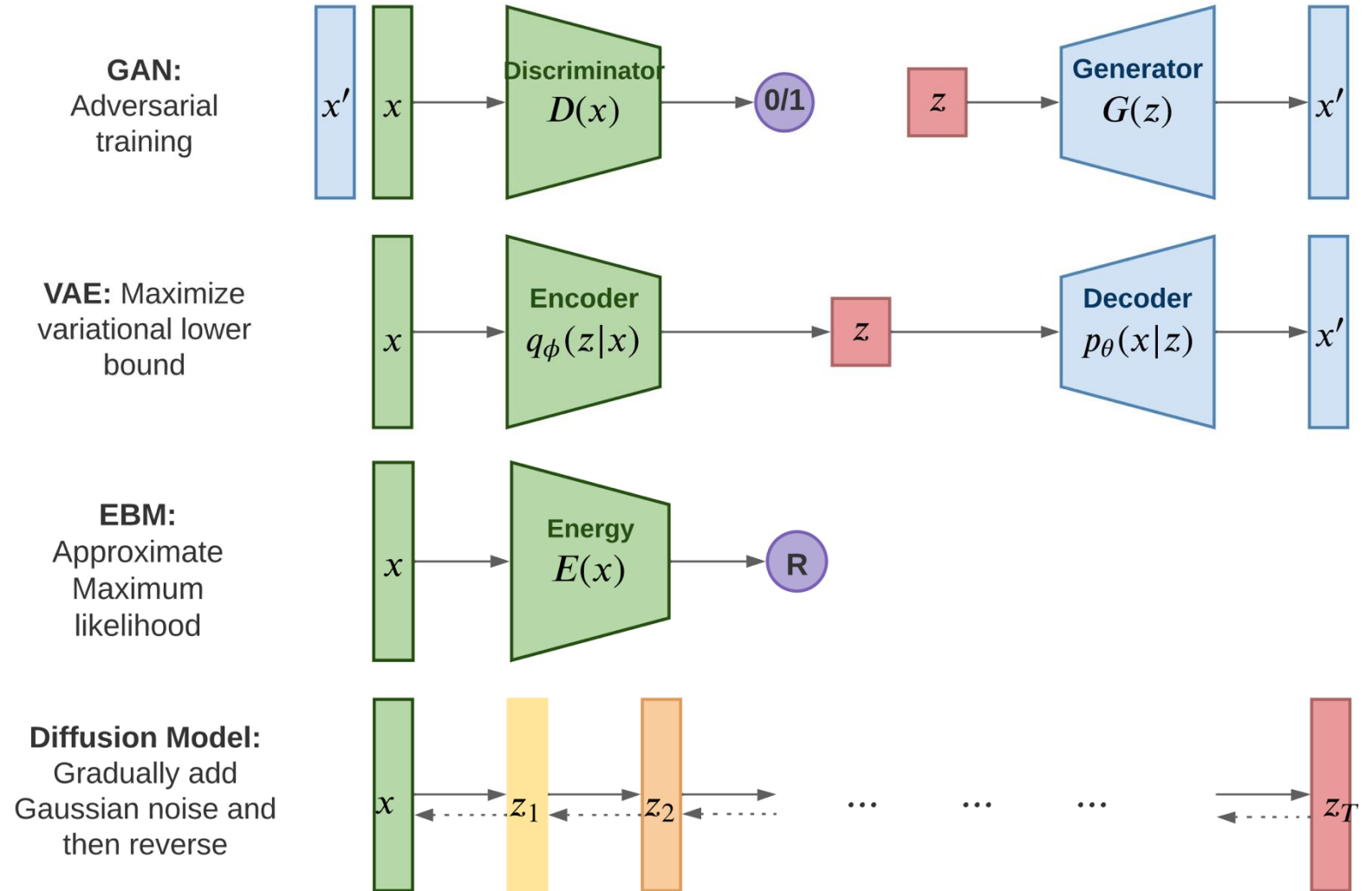
Notation: t – time. $o_{<t}$ - all observations before t . a_t - action at t .

Goal of This Lecture

Use machine learning to build a $p(\text{vision, language})$

Generative Models

- A generative model is a distribution $p(\mathbf{x})$
- Primary kinds: VAE, GAN, EBM, Diffusion



Generative Model Evaluations

- The “standard” way to evaluate a model is to look at the KL divergence against a true distribution

$$D_{\text{KL}}(p \parallel q) = \int p(\mathbf{x}) \log \frac{p(\mathbf{x})}{q(\mathbf{x})}$$

- In language model, the log likelihood can be computed
- Unfortunately, for many image-space models, it is too expensive to compute the likelihood

Likelihood $\neq$ Generation Quality

- Say we have two models: q_0 (good) and q_1 (bad)

$$q_2(\mathbf{x}) = 0.01q_0(\mathbf{x}) + 0.99q_1(\mathbf{x})$$

- $\log q_2(x) \approx \log q_0(x) - 2$
- However, $\log q_0(x)$ is proportional to the dimension

Generative Model Evaluations

- Sample quality
- Sample diversity
- Generalization

Generative Model Evaluations

- **Sample quality**
- **Sample diversity**
- Generalization

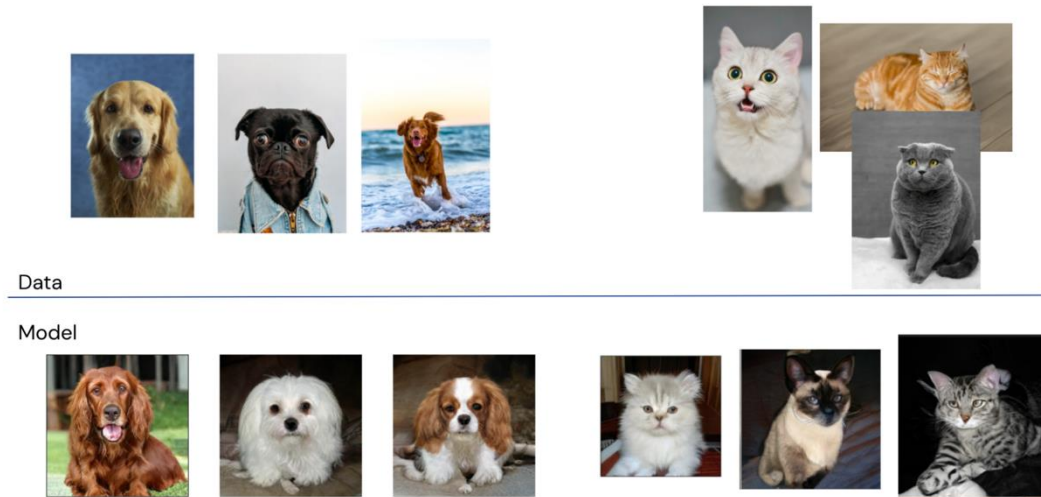
Sample Quality and Diversity

- The basic tool: FID score
- m : samples from the model
- d : samples from the dataset

$$\text{FID} = \|\boldsymbol{\mu}_m - \boldsymbol{\mu}_d\|_2^2 + \text{tr}\left(\boldsymbol{\Sigma}_d + \boldsymbol{\Sigma}_m - 2(\boldsymbol{\Sigma}_d \boldsymbol{\Sigma}_m)^{1/2}\right)$$

- Idea: computing the distribution of model outputs and dataset samples in a latent space (e.g., neural network feature)

Sample Quality and Diversity: FID Mixes the Two



- A bad FID score could mean: model generation is low-quality or model generation is not diverse
- Some evaluation tricks: use nearest neighbor matching to compute “precision” and “recall”

Today's Agenda

Part 1: Principles for T2I Generative Models

- VAEs, GANs, EBMs, Diffusion Models

Part 2: Principles for Implementations of T2I Generative Models

- Classifier-free guidance, label conditions

Today's Agenda

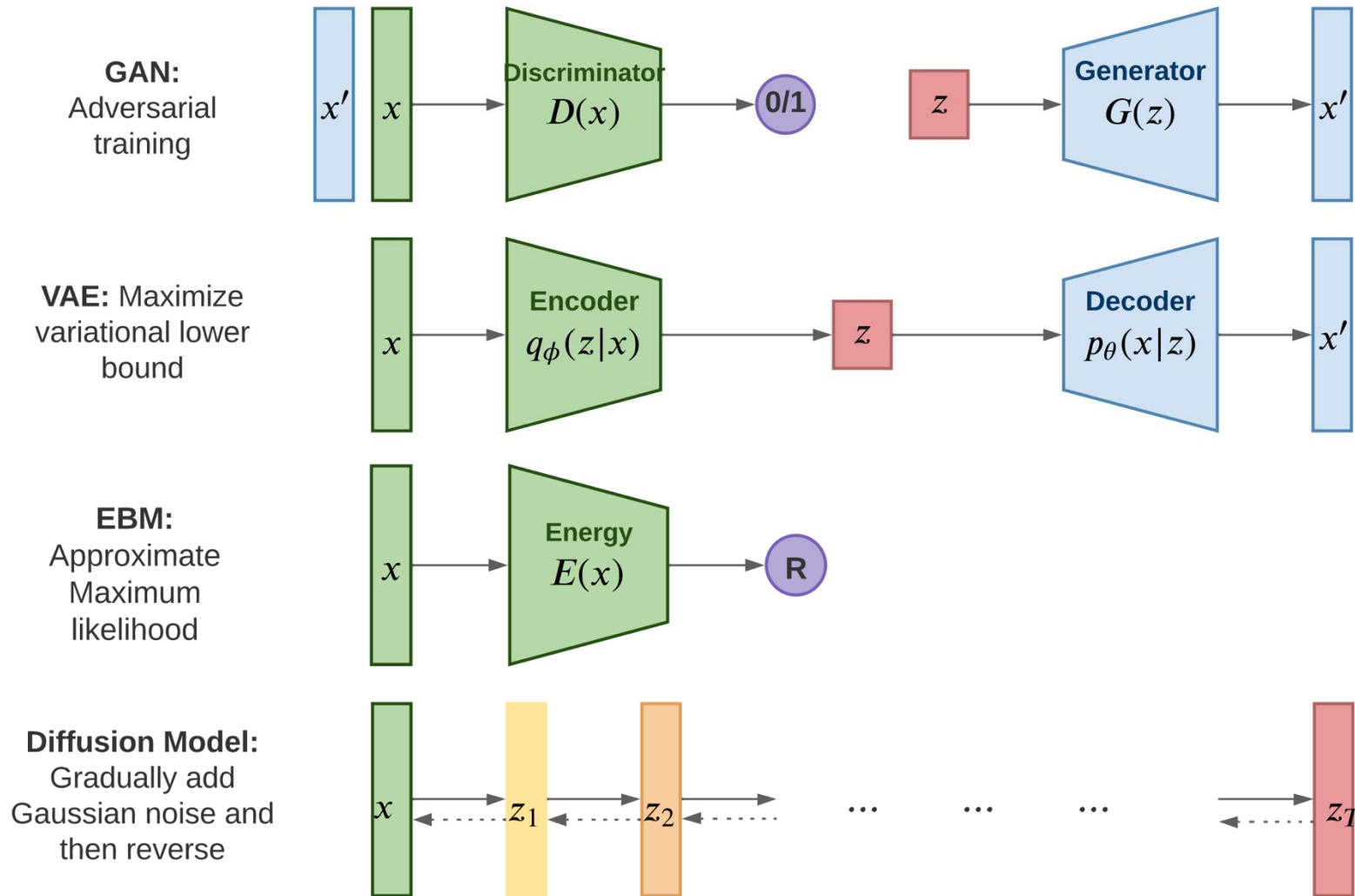
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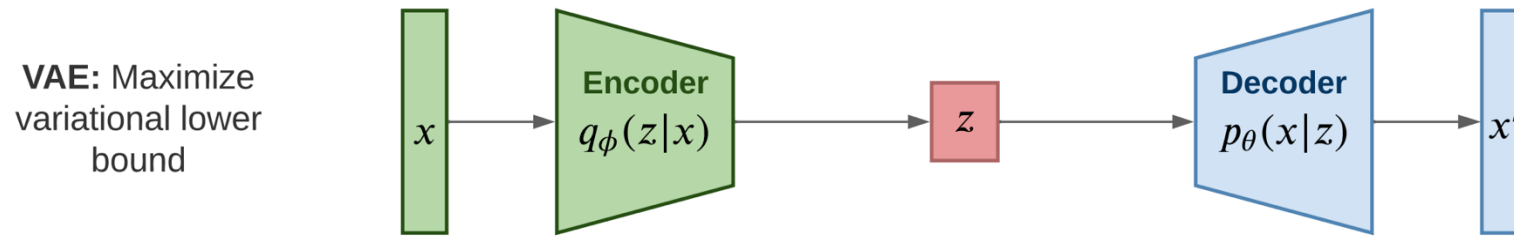
Principles of Text-To-Image Generation (Model)



Principles of Text-To-Image Generation (Model)

- **Generator:** $p_{\theta}(x \mid z)$, producing observations.
- **Energy model:** $\pi_{\alpha}(x) \propto e^{-E_{\alpha}(x)}$, evaluating observations.
- **Inference model:** $q_{\phi}(z \mid x)$, recovering latent explanations.

Principles of Text-To-Image Generation (Model)

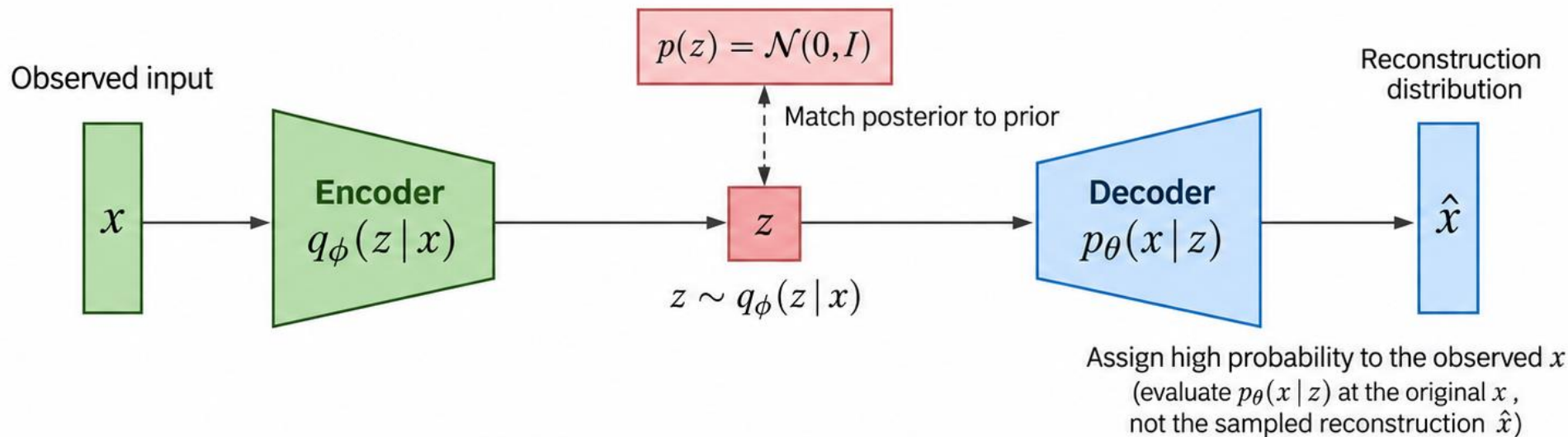


- A VAE model has a generator and a inference model

$$\mathcal{L}_{\beta}(\boldsymbol{\theta}, \phi | \boldsymbol{x}) = \underbrace{-\mathbb{E}_{q_{\phi}(\boldsymbol{z} | \boldsymbol{x})} [\log p_{\boldsymbol{\theta}}(\boldsymbol{x} | \boldsymbol{z})]}_{\mathcal{L}_E} + \beta \underbrace{D_{\text{KL}}(q_{\phi}(\boldsymbol{z} | \boldsymbol{x}) \parallel p_{\boldsymbol{\theta}}(\boldsymbol{z}))}_{\mathcal{L}_R}$$

- There are two views of this loss function

The ELBO View



$$\text{ELBO}(x) = \underbrace{\mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)]}_{\text{Reconstruction: explain the observed } x} - \underbrace{\text{KL}(q_\phi(z|x) || p(z))}_{\text{Regularization: stay close to the prior}}$$

$$\log p_\theta(x) = \text{ELBO}(x) + \underbrace{\text{KL}(q_\phi(z|x) || p_\theta(z|x))}_{\text{Nonnegative posterior gap}}$$

$$\text{Maximize } \text{ELBO}(x) \leq \log p_\theta(x)$$

The Continuous Entropy View

- Goal: build a codec for images that “minimizes” the entropy
- But the codec here is a continuous variable: how to define entropy?
- Solution: define a measure

$$D(p \parallel m) = \int p(x) \log \frac{p(x)}{m(x)} dx.$$

Information-theoretic view: how many additional bits?

$$R(x) = \frac{\text{KL}(q_\phi(z \mid x) \parallel p(z))}{\ln 2}$$

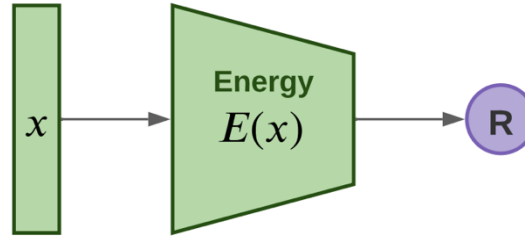
Expected extra bits for coding z with the Gaussian prior instead of the posterior.

Two Views of the Loss Function

- Both views are useful: one is from statistical ML, the other one is from information-theoretic optimization
- The second view yields VAE generalizations such as beta-VAE

Energy-Based Models

EBM:
Approximate
Maximum
likelihood



- Let's define the density function directly

$$p_{\theta}(\mathbf{x}) = \frac{\exp(-\mathcal{E}_{\theta}(\mathbf{x}))}{Z_{\theta}}$$

- Z is known as the partition function

$$Z_{\theta} = \int \exp(-\mathcal{E}_{\theta}(\mathbf{x})) \, d\mathbf{x}$$

- Key insight: We can draw samples from p without computing Z

Langevin Dynamics (MCMC)

$$p_{\theta}(\mathbf{x}) = \frac{\exp(-\mathcal{E}_{\theta}(\mathbf{x}))}{Z_{\theta}}$$

- Intuition: let's do gradient decent on this function to get a better and better sample

$$\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) = -\nabla_{\mathbf{x}} \mathcal{E}_{\theta}(\mathbf{x}) - \underbrace{\nabla_{\mathbf{x}} \log Z_{\theta}}_{=0} = -\nabla_{\mathbf{x}} \mathcal{E}_{\theta}(\mathbf{x}).$$

$$\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \frac{\epsilon^2}{2} \underbrace{\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}^k)}_{=-\nabla_{\mathbf{x}} \mathcal{E}_{\theta}(\mathbf{x})} + \epsilon \mathbf{z}^k, \quad k = 0, 1, \dots, K-1.$$

- $\mathbf{z}^k$ are Gaussian noises. Why do we need ϵ^2 ?

The “Correct” Noise Schedule

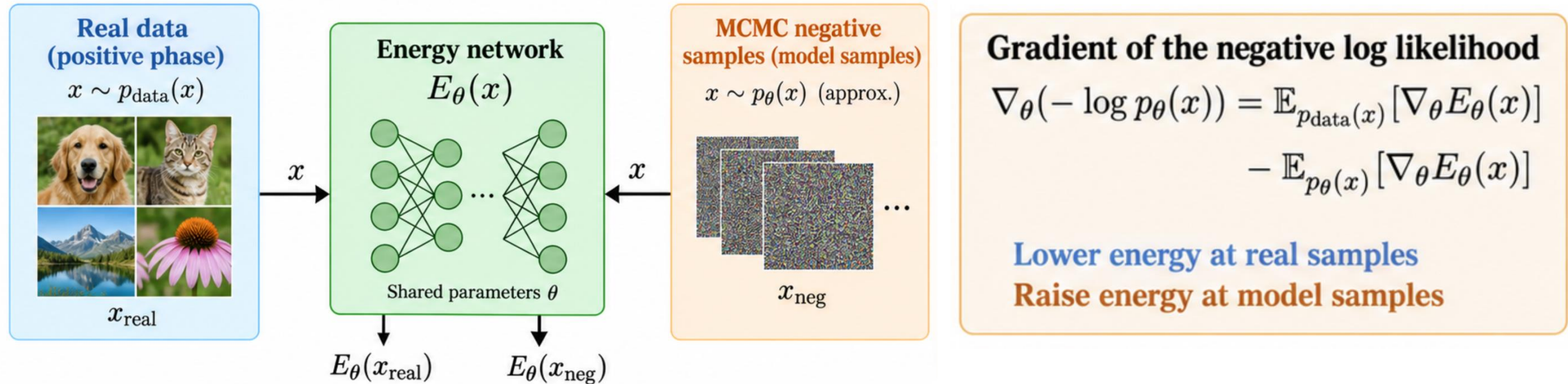
- $\mathbf{z}^k$ are Gaussian noises. Why do we need ϵ^2 ?

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- What’s the variance of $\sum \epsilon \mathbf{z}^k$?
- ϵ^2 !
- We require that the total noise steps $>$ total gradient steps to avoid local optima

Optimization of Energy-Based Models

- First idea: contrastive learning



- You can do other contrastive losses too (hard-negative sampling, softmax)
- We are getting close to GANs :)

Score-Based Models

- We need to know the gradient of ε anyway

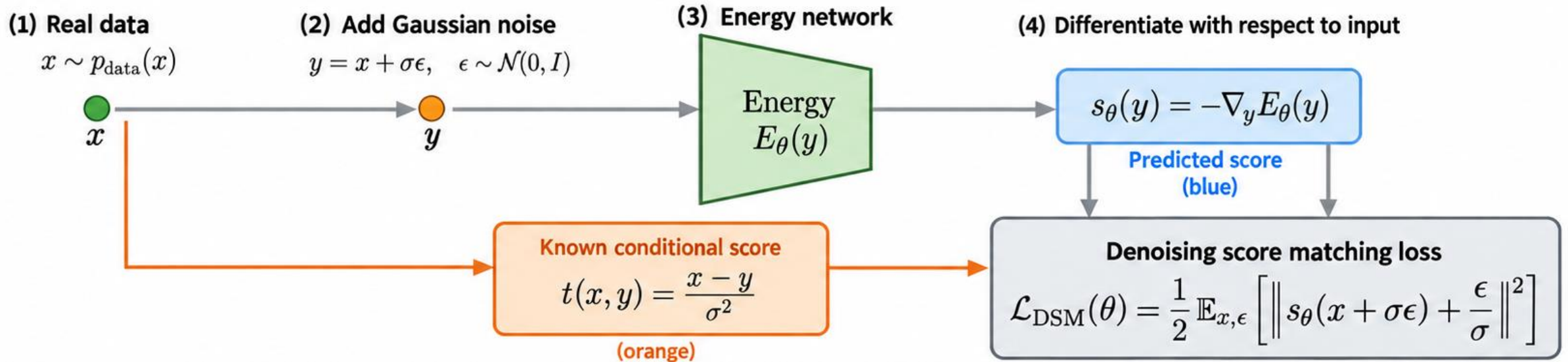
$$\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \frac{\epsilon^2}{2} \underbrace{\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}^k)}_{= -\nabla_{\mathbf{x}} \mathcal{E}_{\boldsymbol{\theta}}(\mathbf{x})} + \epsilon \mathbf{z}^k, \quad k = 0, 1, \dots, K - 1.$$

- Let's just learn the $\nabla_x \varepsilon(x)$ directly
- This is usually called the score function

Optimization of Energy-Based Models

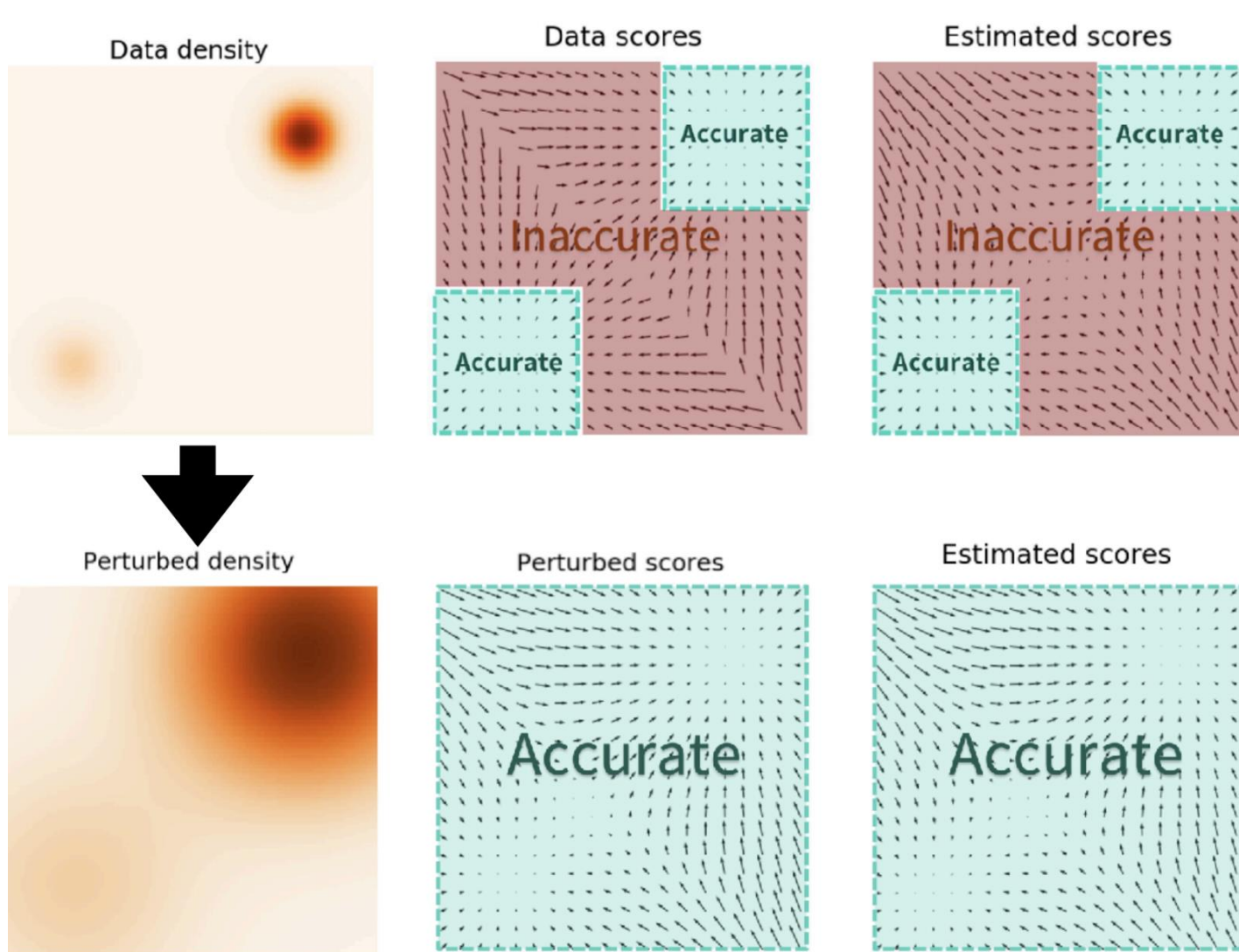
- Second idea: **Score Matching**
- Key insight: If two energy functions have the same first-order derivative, they are the same

TRAINING



- We are getting close to diffusion :)

Noise Choice in Score Matching



$$\mathbb{E}_{p(\mathbf{x})}[\|\nabla_{\mathbf{x}} \log p(\mathbf{x}) - \mathbf{s}_{\theta}(\mathbf{x})\|_2^2] = \int p(\mathbf{x}) \|\nabla_{\mathbf{x}} \log p(\mathbf{x}) - \mathbf{s}_{\theta}(\mathbf{x})\|_2^2 d\mathbf{x}.$$

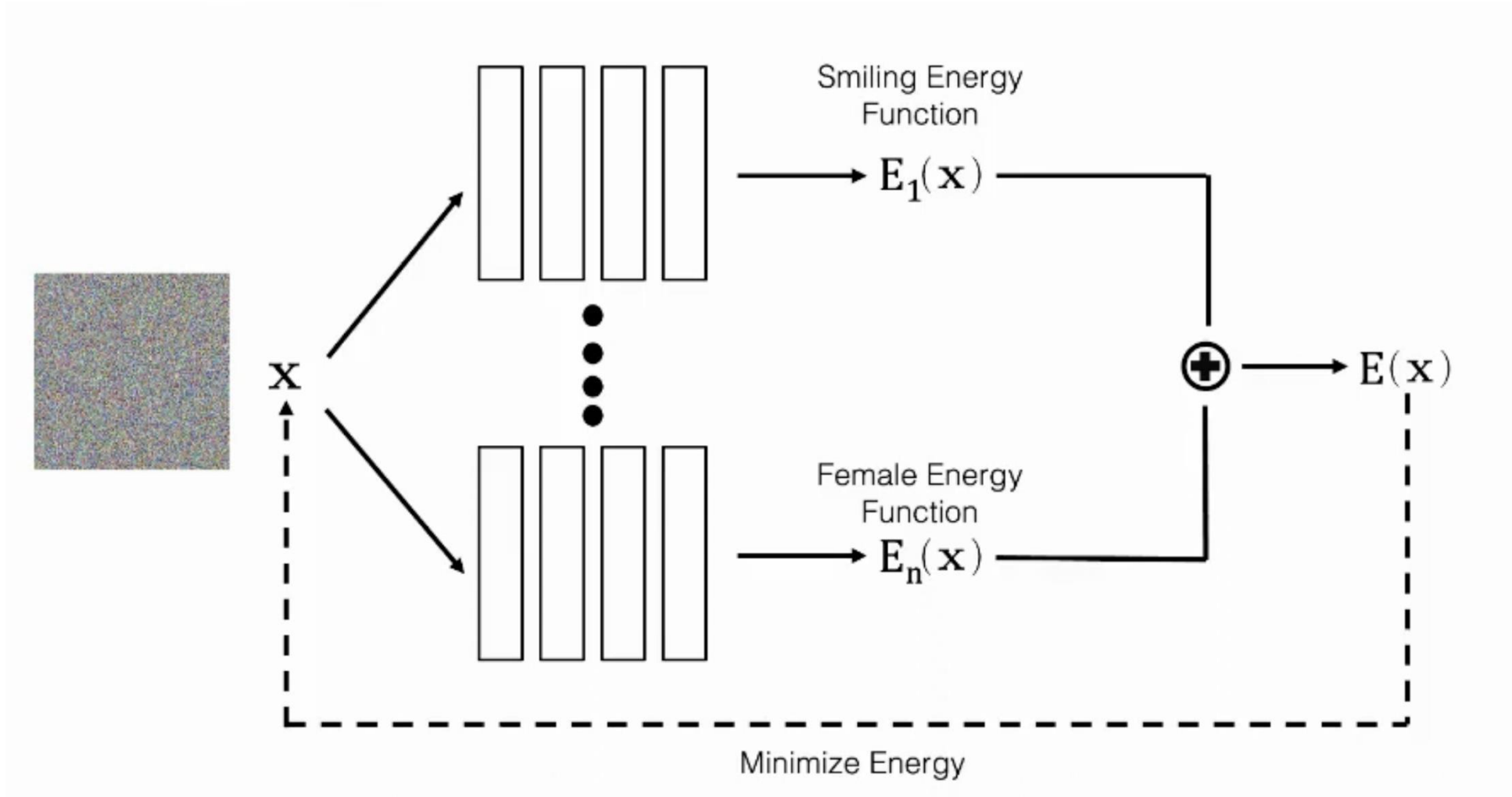
↑

$$p_{\sigma_i}(\mathbf{x}) = \int p(\mathbf{y}) \mathcal{N}(\mathbf{x}; \mathbf{y}, \sigma_i^2 I) d\mathbf{y}.$$

↑
Large noise

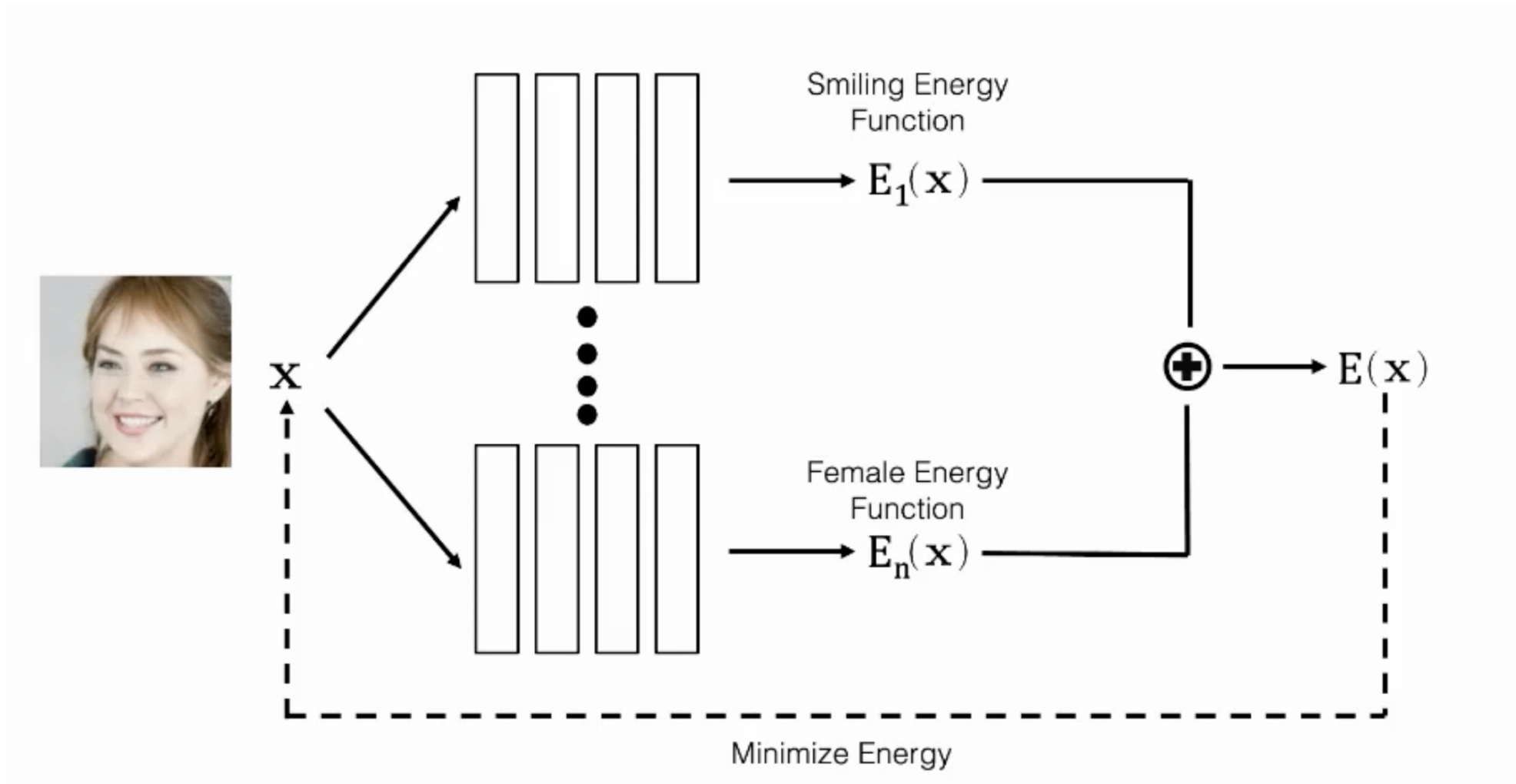
Good Properties of Energy-Based Models

- Energies are Additive



Good Properties of Energy-Based Models

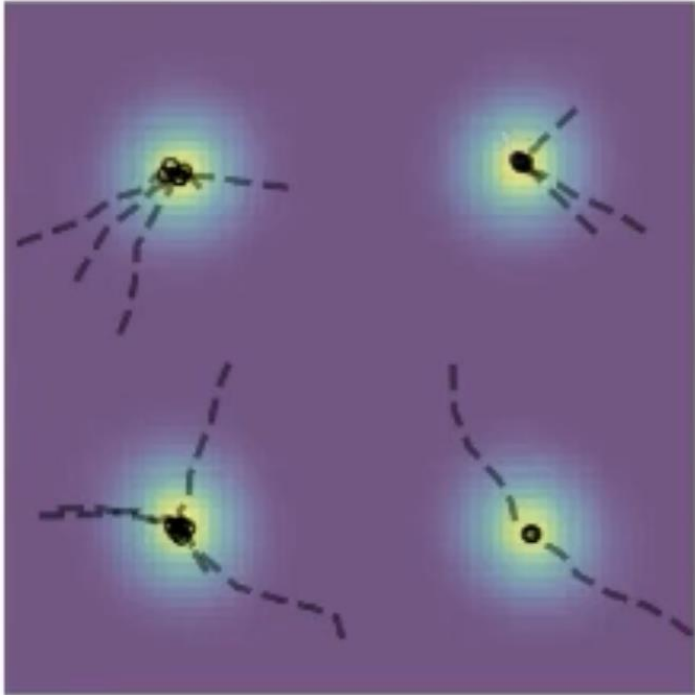
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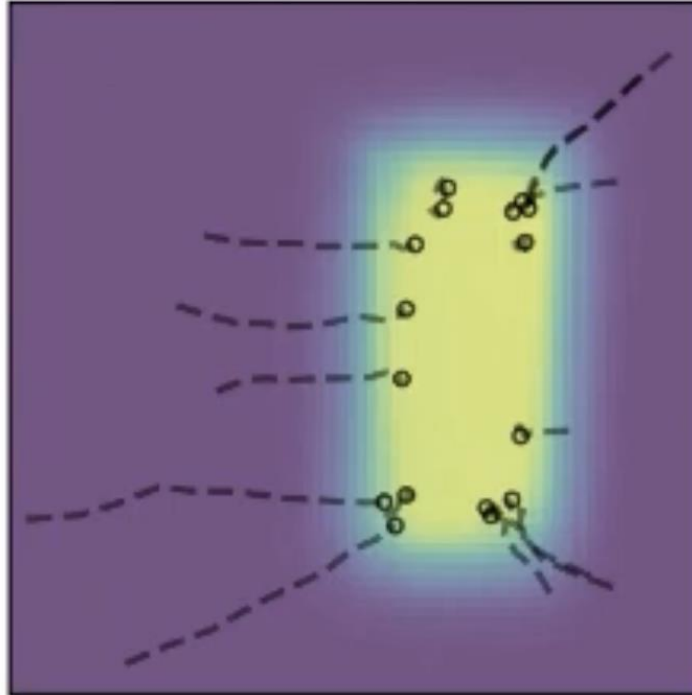
Good Properties of Energy-Based Models

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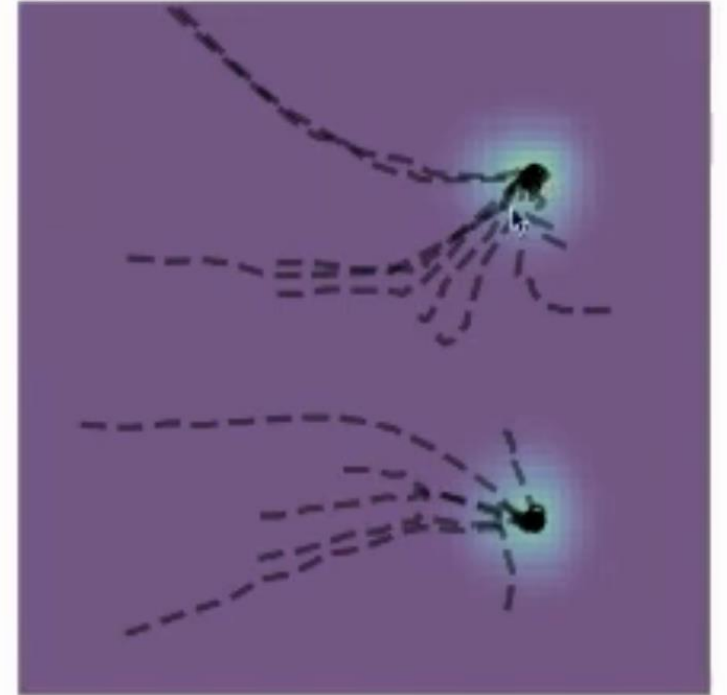
energy A



energy B

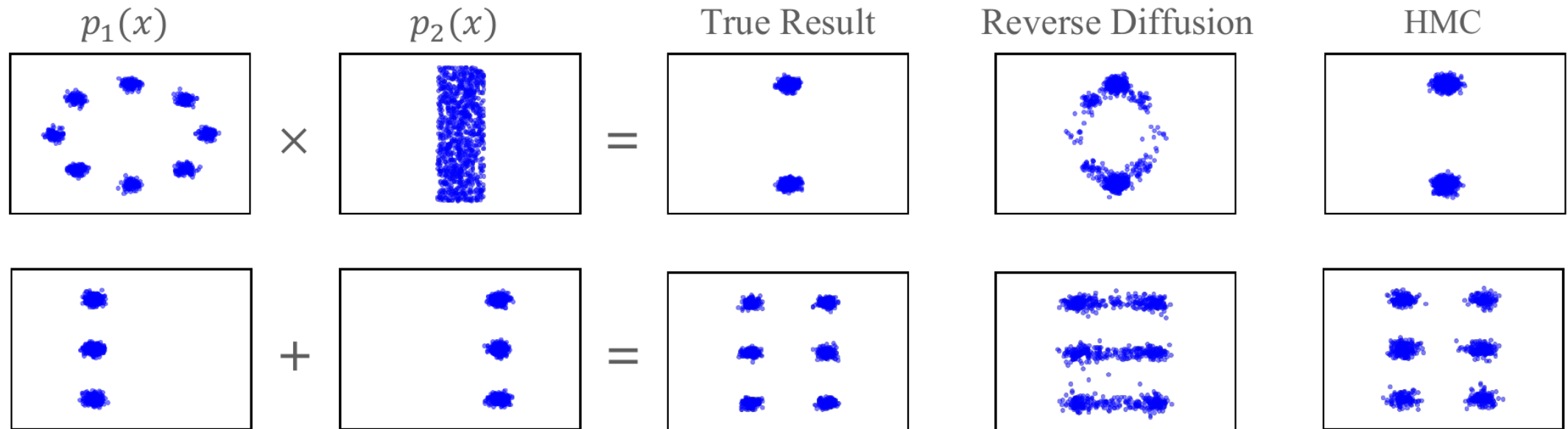


energy A + B



Caveats about Energy Composition

- Directly adding up the score = an MCMC estimate of the true score
- It is NOT a closed-form solution to the joint score
- Therefore, doing 1-step MCMC does not guarantee convergence



Good Properties of Energy-Based Models

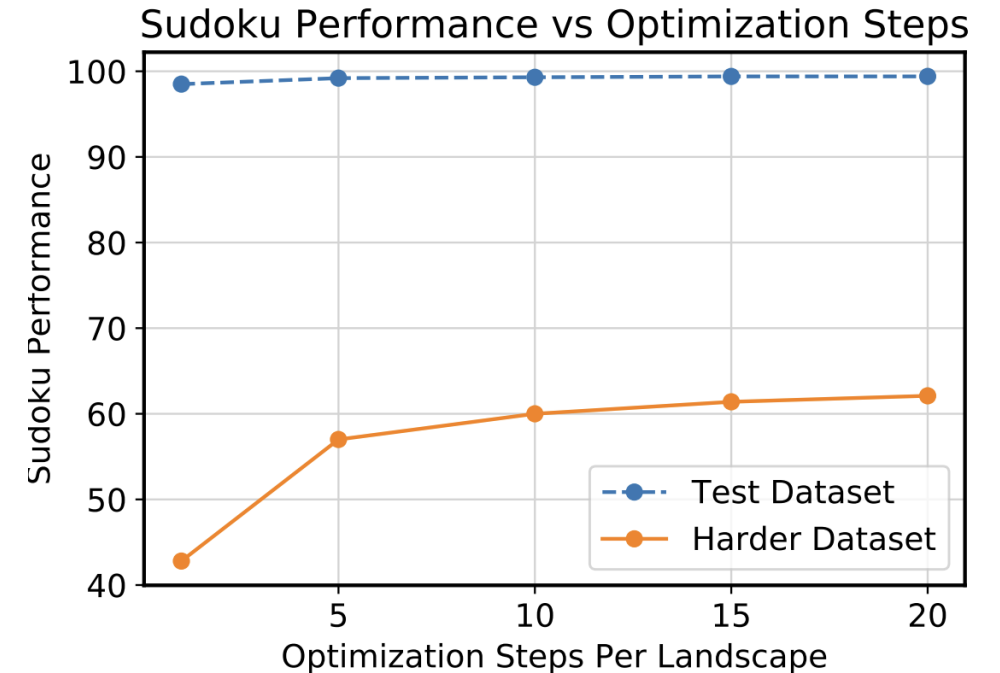
- With energies, we can do adaptive computation

		8		2	1			
5			4					
	4			9	8			6
6		1		5	9	2		4
	8		3	1	2		7	
2		9	8	4		1		5
9			1	7			2	
					4			7
			2	6		9		

Sudoku

		6	3	8				
9				4		3		7
	2							
	5	1					9	
			7	9	2	5		
		6					3	
							8	
	3		9		1			7
				8			1	

(Few Given Numbers)



- With more optimization steps, we can solve harder problems

Good Properties of Energy-Based Models

- With energies, we can do adaptive computation

		8		2	1	
5			4			
	4			9	8	
6		1		5	9	2
	8		3	1	2	
2		9	8	4		1
9			1	7		
					4	
			2	6		9

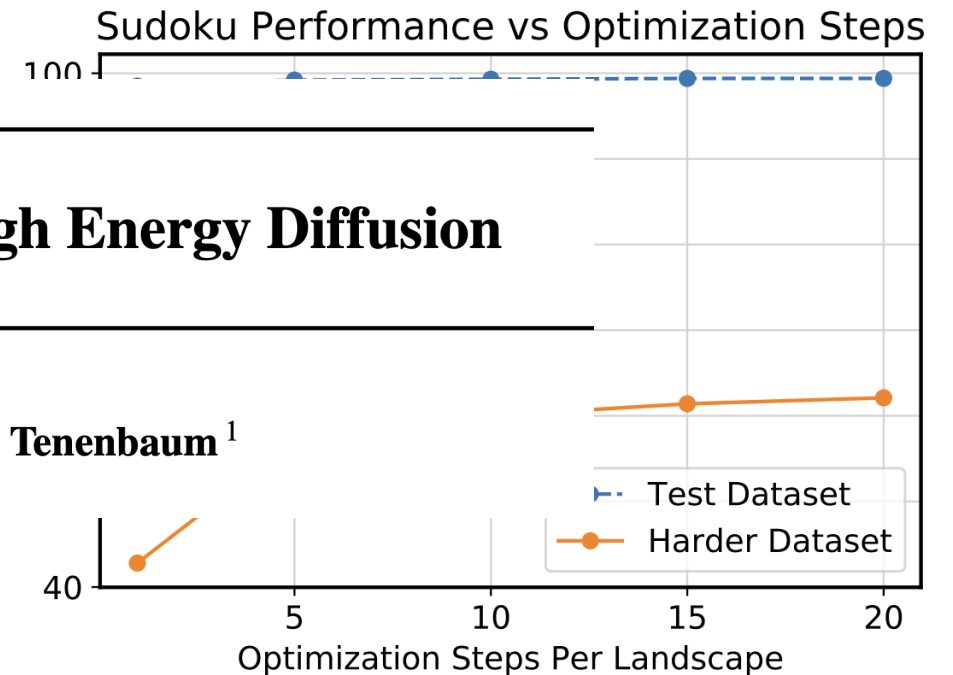
Sudoku

6 3 8

Learning Iterative Reasoning through Energy Diffusion

Yilun Du^{1*} Jiayuan Mao^{1*} Joshua Tenenbaum¹

(Few Given Numbers)



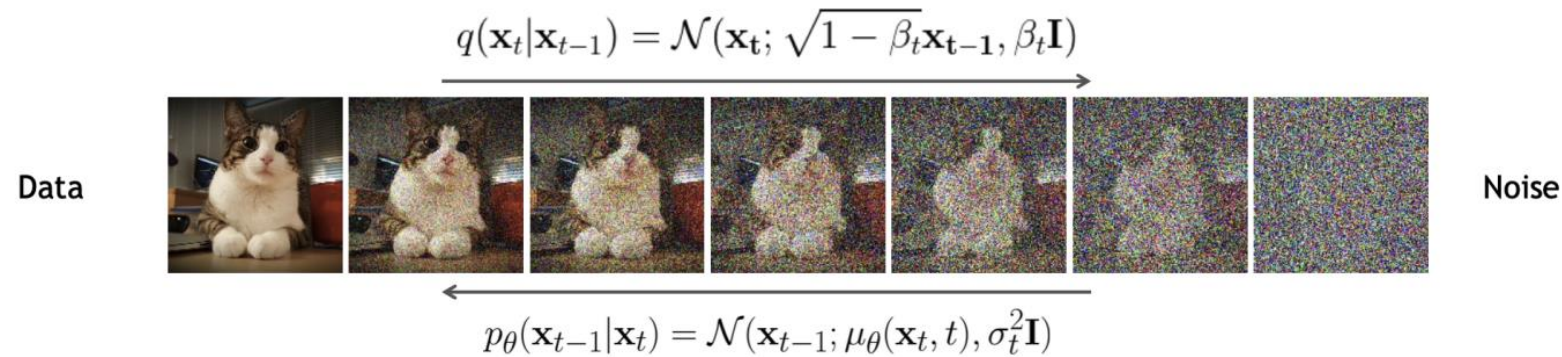
- With more optimization steps, we can solve harder problems

Diffusion Models

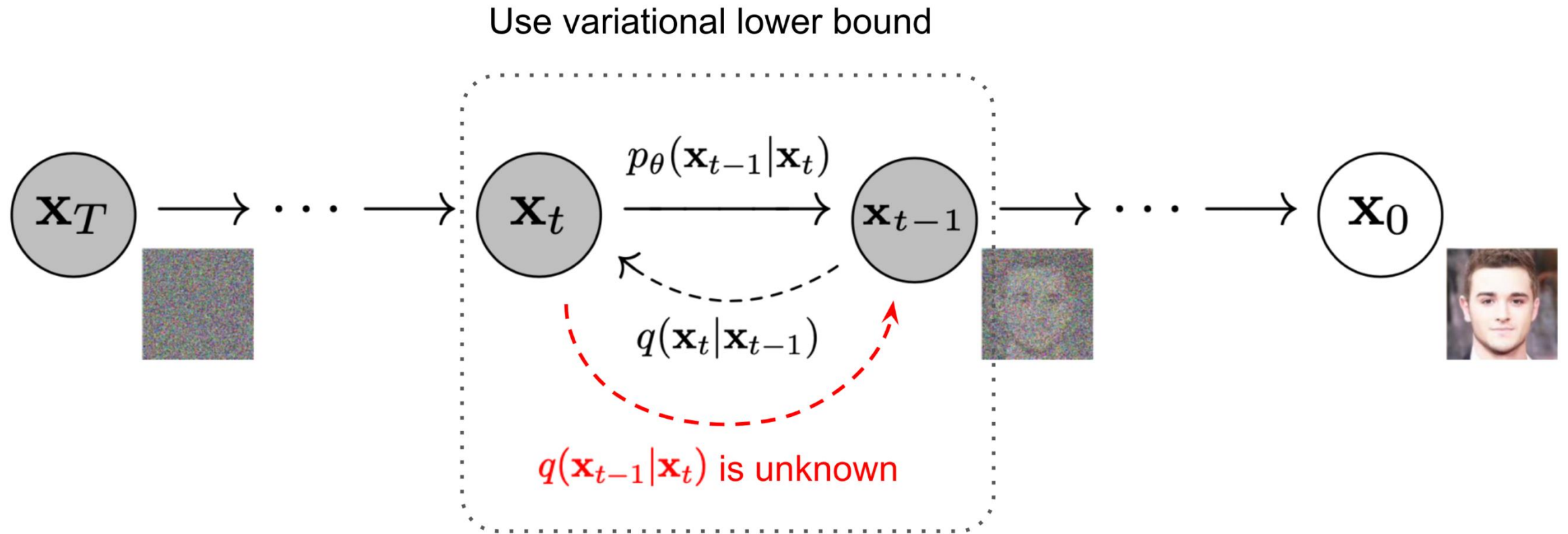
Diffusion Model:
Gradually add
Gaussian noise and
then reverse



- There are so many good resources online
- <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>



Diffusion Models



Diffusion Models

Algorithm 1 Training

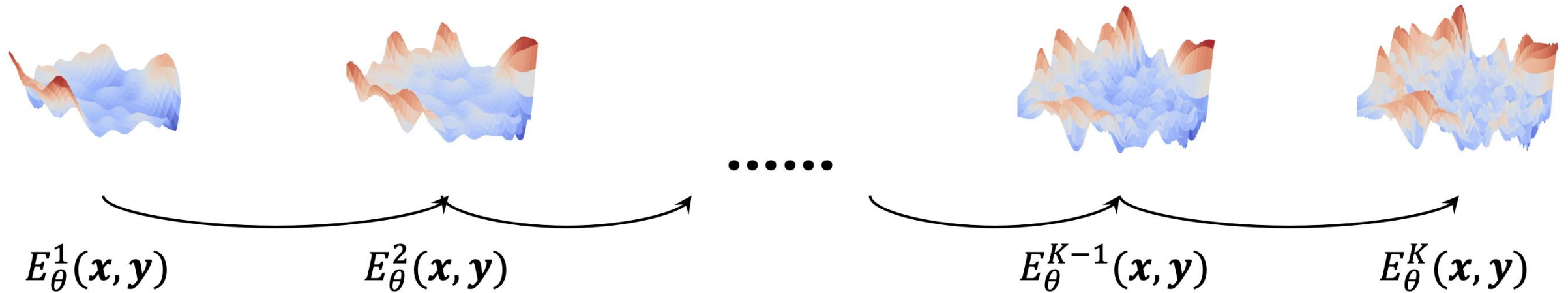
```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
        $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t)\|^2$   
6: until converged
```

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
2: for  $t = T, \dots, 1$  do  
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$   
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$   
5: end for  
6: return  $\mathbf{x}_0$ 
```

Diffusion Model: Notes

- Diffusion models can be viewed as a sequence of EBMs
- Between different stages, we do only 1 step of Langevin dynamics



- It is easy to train because we are supervising on the score function!

Principles of Generative Models (for V-L)

- Key ideas of modeling: generator, inference model, and energy model
- Key ideas of optimizing:
 - MLE, with ELBO (VAEs, diffusion models)
 - negative contrastive learning (GAN, EBMs)
 - score matching (EBMs, diffusion models)
- Key advantages of the EBM view: compositionality, iterated optimization
- Many ideas can mix and match

Today's Agenda

Part 1: Principles for T2I Generative Models

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Part 2: Principles for Implementations of T2I Generative Models

- Classifier-free guidance, label conditions

BigGAN

- Standard GAN-Training with a Hinge loss

$$L_D = \mathbb{E}_{x,y}[\max(0, 1 - D(x, y))] + \mathbb{E}_{z,y}[\max(0, 1 + D(G(z, y), y))]$$

$$L_G = -\mathbb{E}_{z,y}[D(G(z, y), y)]$$

BigGAN

- Using a larger batch size gives more precise gradient estimation
- GAN training requires a lot of tricks like this
- They also mentioned spectral normalization

Batch size	Channels	Parameters	FID ↓	IS ↑
256	64	81.5M	18.65	52.52
512	64	81.5M	15.30	58.77
1,024	64	81.5M	14.88	63.03
2,048	64	81.5M	12.39	76.85

Important Trick: Truncation

- During sampling, only generate from latents close to the mean



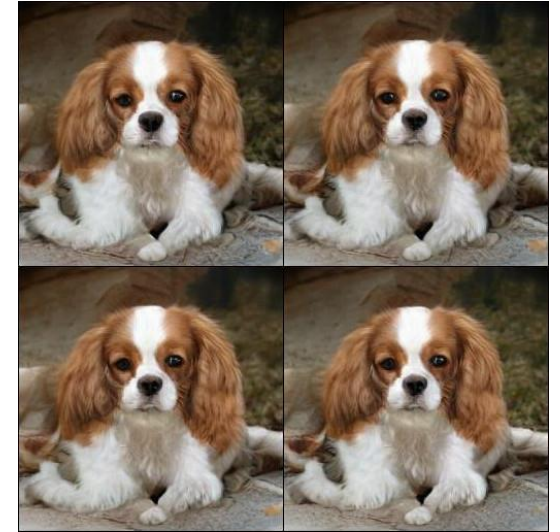
Threshold 2



Threshold 1



Threshold 0.5



Threshold 0.04

DDPM Practice: Prediction Targets

- Noise prediction:

$$\hat{\epsilon} = \epsilon_{\theta}(x_t, t)$$

$$L_{\epsilon} = \mathbb{E}[\|\hat{\epsilon} - \epsilon\|^2]$$

$$\hat{x}_0 = \frac{x_t - b_t \hat{\epsilon}}{a_t}$$

- Target prediction:

$$\hat{x}_0 = x_{0,\theta}(x_t, t)$$

$$L_{x_0} = \mathbb{E}[\|\hat{x}_0 - x_0\|^2]$$

- Velocity Prediction:

$$v = x_0 - \epsilon$$

$$\hat{v} = v_{\theta}(z_t, t)$$

$$L_v = \mathbb{E}[\|\hat{v} - v\|^2]$$

$$\hat{x}_0 = z_t + (1 - t)\hat{v}$$

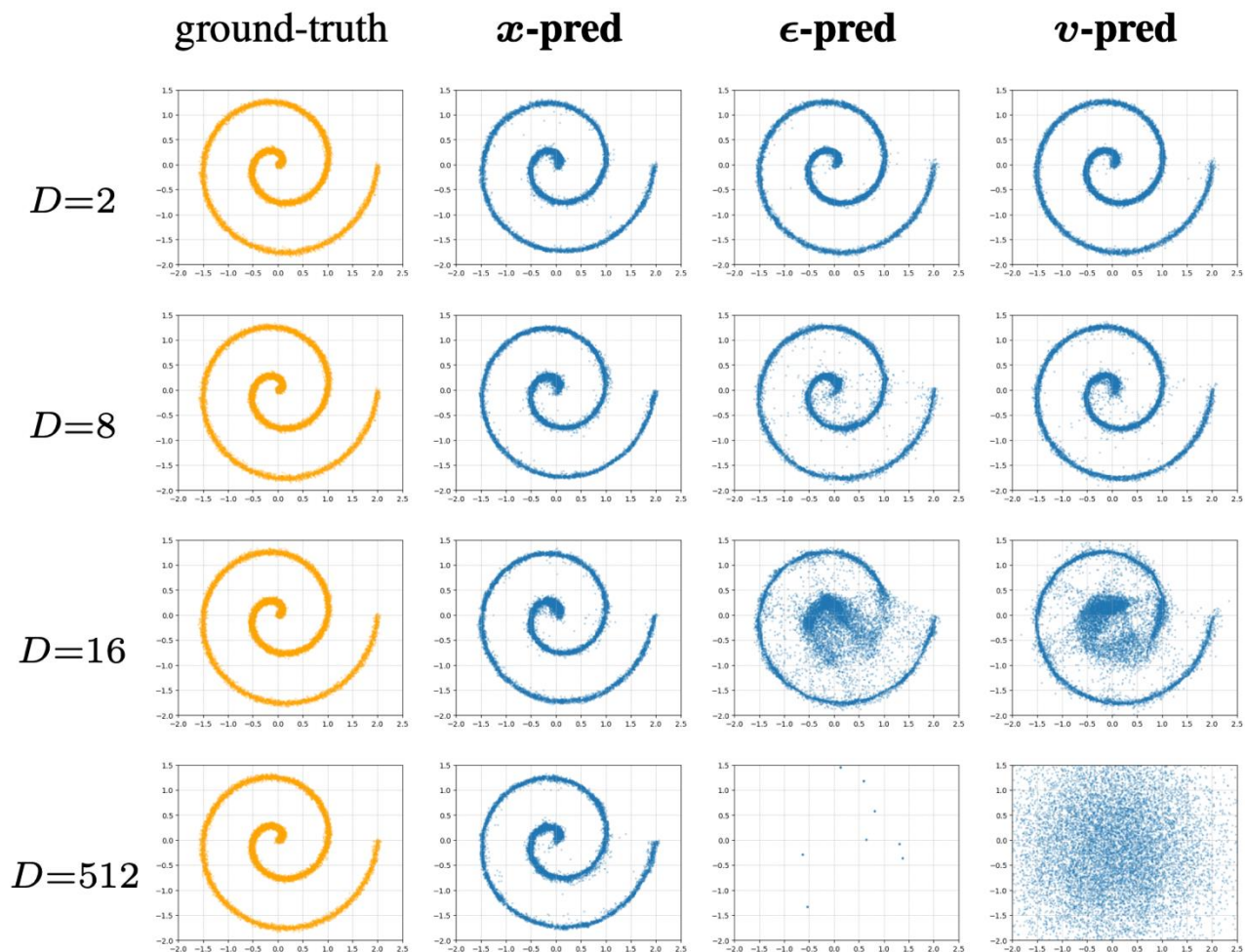
$$\hat{\epsilon} = z_t - t\hat{v}$$

DDPM Practice: Prediction Targets

	(a) x -pred	(b) ϵ -pred	(c) v -pred
	$\mathbf{x}_\theta := \text{net}_\theta(\mathbf{z}_t, t)$	$\boldsymbol{\epsilon}_\theta := \text{net}_\theta(\mathbf{z}_t, t)$	$\mathbf{v}_\theta := \text{net}_\theta(\mathbf{z}_t, t)$
(1) x -loss: $\mathbb{E} \ \mathbf{x}_\theta - \mathbf{x}\ ^2$	$\mathbf{x}_\theta$	$\mathbf{x}_\theta = (\mathbf{z}_t - (1-t)\boldsymbol{\epsilon}_\theta)/t$	$\mathbf{x}_\theta = (1-t)\mathbf{v}_\theta + \mathbf{z}_t$
(2) ϵ -loss: $\mathbb{E} \ \boldsymbol{\epsilon}_\theta - \boldsymbol{\epsilon}\ ^2$	$\boldsymbol{\epsilon}_\theta = (\mathbf{z}_t - t\mathbf{x}_\theta)/(1-t)$	$\boldsymbol{\epsilon}_\theta$	$\boldsymbol{\epsilon}_\theta = \mathbf{z}_t - t\mathbf{v}_\theta$
(3) v -loss: $\mathbb{E} \ \mathbf{v}_\theta - \mathbf{v}\ ^2$	$\mathbf{v}_\theta = (\mathbf{x}_\theta - \mathbf{z}_t)/(1-t)$	$\mathbf{v}_\theta = (\mathbf{z}_t - \boldsymbol{\epsilon}_\theta)/t$	$\mathbf{v}_\theta$

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DDPM Practice: Prediction Targets

	(a) x -pred	(b) ϵ -pred	(c) v -pred
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Back to Basics: Let Denoising Generative Models Denoise

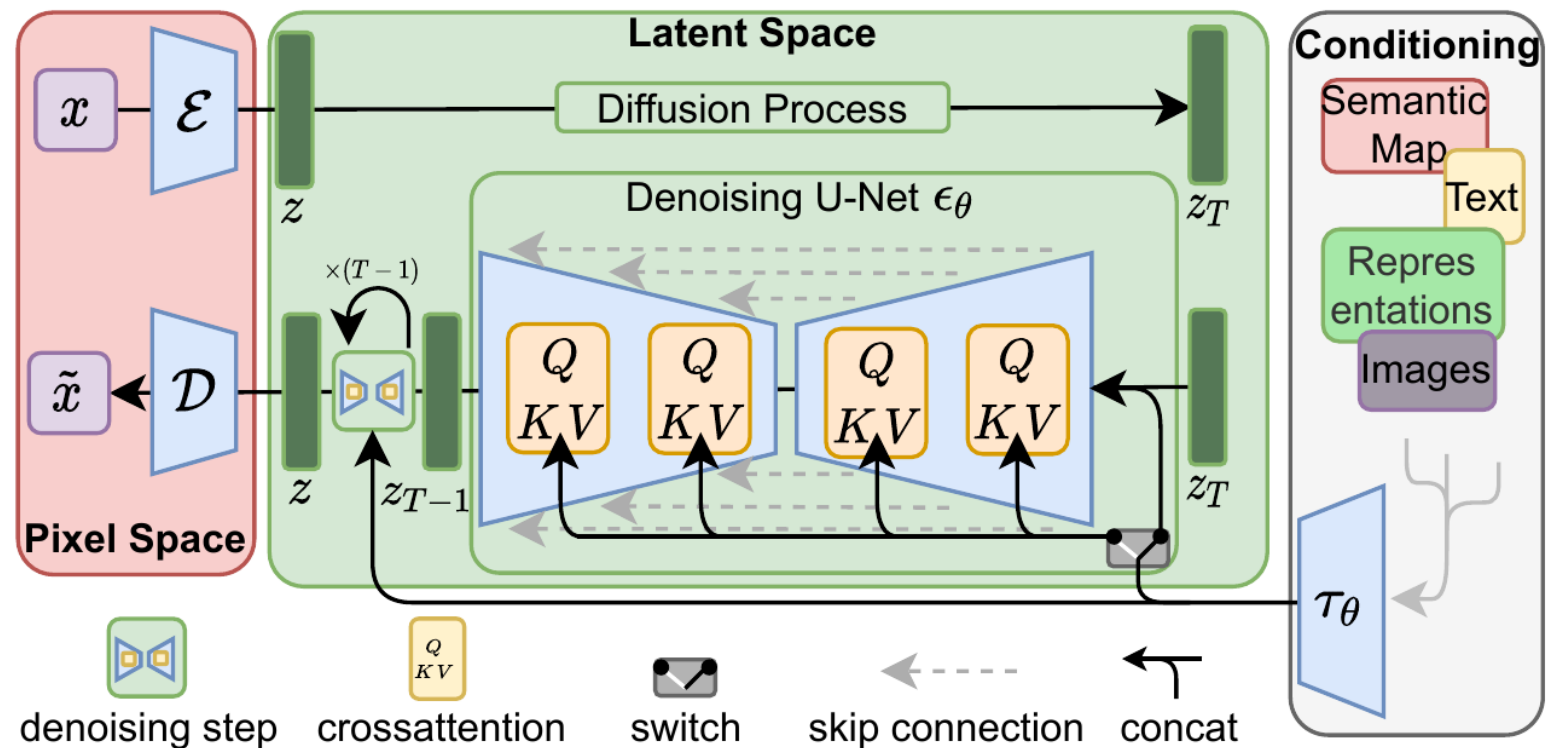
Tianhong Li Kaiming He

MIT

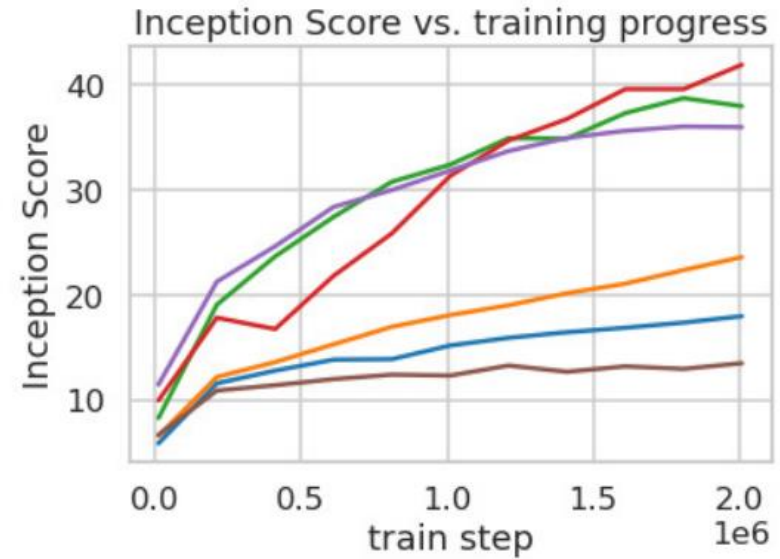
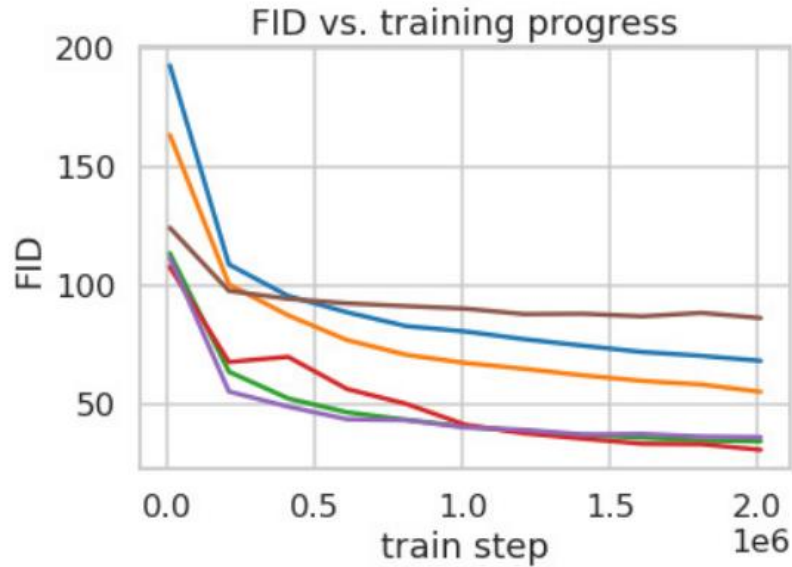
- Conclusion: Use x -prediction, v -loss
- Intuition: CNNs are good for modeling the “real” image space

Latent Diffusion

- The diffusion acts on spatial latent features instead of RGB pixels
- At generation time, sample a latent and decode once; Text enters the denoiser through cross-attention



Compression Ratio



- Very small down sampling leaves expensive image modeling to diffusion
- In practice, all of these LDMs are very “local;” Changing a single code dimension introduces small local changes in the image

Cross-Attention for Vision-Language Alignment

$$Q = HW_Q, \quad K = TW_K, \quad V = TW_V$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

- This is similar to the MaskCLIP alignment trick between language latents and image latents

Classifier Guidance

$$\nabla_{x_t} \log p_t(x_t | c) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log p_t(c | x_t)$$

- Goal: use a classifier to guide the model to generate a particular class
- Issue: need a classifier that works for every noise level

Classifier Guidance: Dropout Training

$$\nabla_{x_t} \log p_t(x_t | c) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log p_t(c | x_t)$$

- Sample an image–condition pair
- Replace the condition with a null condition with probability p
- Train the same denoiser with the same noise-prediction loss

Classifier Guidance: Inference

$$\hat{\epsilon} = \epsilon_u + s(\epsilon_c - \epsilon_u)$$

- Scale 0 gives the unconditional prediction; scale 1 gives ordinary conditional sampling
- Larger scales emphasize the difference induced by the condition

Classifier Guidance: The Density

$$\hat{s}_t = s_{u,t} + s(s_{c,t} - s_{u,t})$$

$$\hat{s}_t = \nabla \log[p_t(x_t | c)^s p_t(x_t)^{1-s}]$$

$$p_t(x_t | c)^s p_t(x_t)^{1-s} \propto p_t(x_t) p_t(c | x_t)^s$$

- We are basically trading-off diversity and quality
- Similar to the truncation trick!

Higher Guidance Can Worsen FID while Raising IS

Guidance w	Equivalent s	FID ↓	IS ↑
0.0	1.0	7.27	82.45
0.3	1.3	2.43	158.47
1.0	2.0	7.86	297.98
4.0	5.0	21.53	421.03

Higher Guidance Can Worsen FID while Raising IS



Left: ordinary conditional sampling. Right: classifier-free guidance with $w = 3$.

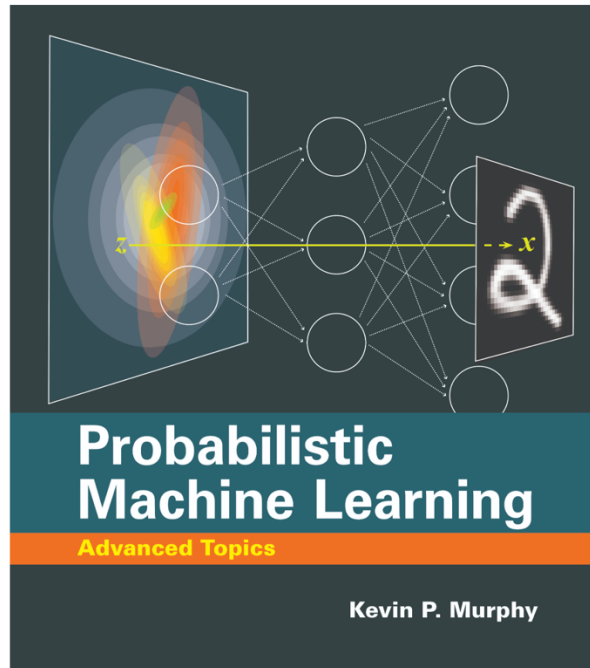
Principles of Implementations of V-L Generative Models

- Key idea 1: condition your generator
- Key idea 2: trade-off diversity and quality (BigGAN, CFG)

Reference Books

Probabilistic Machine Learning: Advanced Topics

by [Kevin Patrick Murphy](#).
MIT Press, 2023.



Questions for Discussion

- What are the tradeoffs between generators, inference models, and energy models?
- What would be a good evaluation metric for generative models in your domain?
- What are compositional structures in your research domain?